Andrés Sànchez

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Full stack engineer (Python/Django, PostgreSQL, JavaScript/React) with 3 years building real-time software: a year on a national-scale precision-irrigation platform at IRTA, and two years shipping four commercial Unity titles for global brands. Strong in performance work, data-heavy map interfaces and AI-assisted workflows, with documented technical decisions.

EXPERIENCE

IRTA | Lleida, Spain **Jan 2026 - Present**

Full-stack developer

- Built a web platform that **helps farmers plan irrigation** using crop, weather and sensor data (Python/Django, PostgreSQL, JavaScript, CSS).
- **Built an AI-assisted translation system** for the interface, with a verified glossary so 490 professionally reviewed translations always override the AI. Localized about 13,000 strings into 6 languages with a local LLM (Ollama, qwen2.5:32b, zero API cost), an estimated 140+ hours of manual work.
- Adapted one codebase to 10+ regional irrigation authorities, each with its own configuration and terminology, without maintaining separate code.
- Led the frontend refactor from a legacy page to a modern map-centered interface (Google Maps style).
- Cut map load time by resolving parcel polygons to the zoom level's point resolution, so far less data is loaded.
- Automated task creation through the GitLab Issues API.

Petoons Studio | Barcelona, Spain **Oct 2023 - Jan 2026**

Lead Porting Engineer - Gameplay Programmer - UI Programmer

- **Shipped four Unity titles** for Android and other platforms featuring world-famous IPs such as PJ Masks, Hot Wheels and Monster High, working with partners such as BMW.
- Profiled and fixed severe memory and framerate issues on Android and console hardware, and advised layout and 3D artists on Unity optimization.
- Led the porting department, integrating **new technologies** such as console devkits, platform-specific APIs and third-party tools as clients required.
- Built the real-time four-player multiplayer mode of Hot Wheels XTreme Overdrive (client-server, steady 30 fps on mobile) and adapted the game to BMW iX3 in-car displays.

PROJECTS

EstalvIA | Python, Django, PostgreSQL, LLM (live demo and repo)

- AI reads supplier delivery notes and invoices, flags what is unpaid or mismatched, and reconciles payments against bank CSV statements using rules instead of AI.
- One Django project mounting each prototype as an app, with per-user data isolation and invite-code registration.
- Decisions recorded as ADRs; tests run without calling the model.

Vibe Composer | JavaScript, React, Supabase, Capacitor (live demo and repo)

- Lyric co-writer for Spanish and Catalan with an AI partner, a React Flow canvas, a Web Audio sequencer and a mobile build with Capacitor.
- Per-user row-level security in Supabase and 120 logic tests with Vitest.

EDUCATION & TECHNICAL SKILLS

Universitat Politècnica de Catalunya - CITM | Barcelona, Spain

B.S. in Videogames Design & Development

Unity: Profiler, memory and performance optimization, asset bundles, multi-platform porting, uGUI, DOTween.

Programming & AI: Python, Django, PostgreSQL, JavaScript, React, Supabase, REST APIs, Git, GitLab; C#, C++; generative AI / LLM workflows

Tools: Git, GitLab, Jira, Jenkins, Bitbucket, Confluence, Android Studio, Microsoft GDK, console target managers.

Languages: English (professional), Spanish, Catalan.